

TCS PLACEMENT PREPARATION

Question Bank 2025

125 Questions | 5 Sections | With Answers & Explanations

Aptitude • Verbal • Reasoning • Technical • Coding

**125
Questions**

**5
Sections**

**Answers &
Explanations**

**NQT
Pattern**

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SECTION 1: QUANTITATIVE APTITUDE

Topics: Averages | Profit & Loss | Time-Speed-Distance | Percentages | Ratios | Work | Interest

Q1.

The average of 5 numbers is 27. If one number is excluded, the average becomes 25. What is the excluded number?

- A) 35
- B) 37
- C) 39
- D) 41

✓ **Answer: A) 35**

Explanation: Total = $5 \times 27 = 135$; remaining $4 \times 25 = 100$; excluded = $135 - 100 = 35$

Q2.

A shopkeeper buys an article for Rs.800 and sells it for Rs.1000. Find the profit percentage.

- A) 20%
- B) 25%
- C) 30%
- D) 15%

✓ **Answer: B) 25%**

Explanation: Profit% = $(200/800) \times 100 = 25\%$

Q3.

A train travels at 60 km/h. How long will it take to cover 240 km?

- A) 3 hours
- B) 4 hours
- C) 5 hours
- D) 6 hours

✓ **Answer: B) 4 hours**

Explanation: Time = Distance/Speed = $240/60 = 4$ hours

Q4.

What is 15% of 480?

- A) 62
- B) 72
- C) 82
- D) 92

✓ **Answer: B) 72**

Explanation: $15/100 \times 480 = 72$

Q5.

The ratio of boys to girls in a class is 3:2. If there are 40 students total, how many are boys?

- A) 20
- B) 24

- C) 28
- D) 30

✓ Answer: B) 24

Explanation: Boys = $3/(3+2) \times 40 = 24$

Q6.

A and B together can complete a work in 12 days. A alone can do it in 20 days. In how many days can B alone complete it?

- A) 25 days
- B) 28 days
- C) 30 days
- D) 32 days

✓ Answer: C) 30 days

Explanation: B's 1-day work = $1/12 - 1/20 = (5-3)/60 = 1/30$, so 30 days

Q7.

Find the simple interest on Rs.5000 at 8% per annum for 3 years.

- A) Rs.1000
- B) Rs.1200
- C) Rs.1500
- D) Rs.1800

✓ Answer: B) Rs.1200

Explanation: $SI = P \times R \times T / 100 = 5000 \times 8 \times 3 / 100 = 1200$

Q8.

If the cost price of 15 articles equals the selling price of 12 articles, find the gain %.

- A) 20%
- B) 25%
- C) 30%
- D) 15%

✓ Answer: B) 25%

Explanation: Gain = $(15-12)/12 \times 100 = 25\%$

Q9.

Two pipes can fill a tank in 10 and 15 hours respectively. How long to fill the tank together?

- A) 4 hours
- B) 5 hours
- C) 6 hours
- D) 7 hours

✓ Answer: C) 6 hours

Explanation: Combined rate = $1/10 + 1/15 = 5/30 = 1/6$, so 6 hours

Q10.

A number when divided by 5 gives remainder 3. What is the remainder when the square of that number is divided by 5?

- A) 1
- B) 2

- C) 3
- D) 4

✓ Answer: D) 4

Explanation: $n=5k+3$; $n^2=(5k+3)^2=25k^2+30k+9$; remainder of $9/5 = 4$

Q11.

Find compound interest on Rs.8000 at 10% p.a. for 2 years.

- A) Rs.1600
- B) Rs.1680
- C) Rs.1760
- D) Rs.1800

✓ Answer: B) Rs.1680

Explanation: $CI = 8000[(1.1)^2 - 1] = 8000*0.21 = 1680$

Q12.

A can do a work in 15 days and B in 20 days. They work together for 5 days, then B leaves. How many more days for A to finish?

- A) 5.25 days
- B) 6 days
- C) 6.25 days
- D) 7 days

✓ Answer: C) 6.25 days

Explanation: Together in 5 days $= 5(1/15+1/20) = 5*7/60 = 7/12$ done. Remaining $= 5/12$. A takes $5/12 \times 15 = 6.25$ days

Q13.

The LCM of 12, 18, and 24 is:

- A) 48
- B) 60
- C) 72
- D) 96

✓ Answer: C) 72

Explanation: $LCM(12,18,24) = 72$

Q14.

If $a:b = 2:3$ and $b:c = 4:5$, find $a:c$.

- A) 8:15
- B) 6:10
- C) 4:9
- D) 2:5

✓ Answer: A) 8:15

Explanation: $a:b=2:3$, $b:c=4:5 \rightarrow a:c = (2 \times 4):(3 \times 5) = 8:15$

Q15.

A sum doubles in 8 years at simple interest. Find the rate of interest.

- A) 10%
- B) 12%

- C) 12.5%
- D) 15%

✓ Answer: C) 12.5%

Explanation: $SI = P$, so $P = P \cdot R \cdot 8 / 100 \rightarrow R = 100 / 8 = 12.5\%$

Q16.

A train 200m long passes a pole in 10 seconds. Find the speed in km/h.

- A) 60
- B) 66
- C) 72
- D) 78

✓ Answer: C) 72

Explanation: $Speed = 200 / 10 = 20 \text{ m/s} = 20 \times 3.6 = 72 \text{ km/h}$

Q17.

The product of two numbers is 120 and their HCF is 6. Find their LCM.

- A) 10
- B) 15
- C) 20
- D) 30

✓ Answer: C) 20

Explanation: $LCM = Product / HCF = 120 / 6 = 20$

Q18.

If 40% of a number is 160, what is 25% of that number?

- A) 80
- B) 90
- C) 100
- D) 110

✓ Answer: C) 100

Explanation: $Number = 160 / 0.4 = 400$; $25\% = 100$

Q19.

A boat travels 30 km upstream in 6 hours and 36 km downstream in 4 hours. Find the speed of the stream.

- A) 1.5 km/h
- B) 2 km/h
- C) 2.5 km/h
- D) 3 km/h

✓ Answer: A) 1.5 km/h

Explanation: $Upstream = 5 \text{ km/h}$, $Downstream = 9 \text{ km/h}$; $stream = (9 - 5) / 2 = 2 \text{ km/h}$... recalc: $(9 - 5) / 2 = 2$. Actually 2 km/h

Q20.

In how many ways can 5 books be arranged on a shelf?

- A) 24
- B) 60

- C) 120
- D) 240

✓ Answer: C) 120

Explanation: $5! = 120$

Q21.

The difference between a two-digit number and the number obtained by reversing its digits is 36. If the sum of digits is 10, find the number.

- A) 73
- B) 82
- C) 91
- D) 64

✓ Answer: A) 73

Explanation: Let digits be x, y : $x+y=10$, $10x+y-(10y+x)=36 \rightarrow 9(x-y)=36 \rightarrow x-y=4$. Solving: $x=7, y=3 \rightarrow 73$

Q22.

What is the probability of getting a prime number when a die is rolled?

- A) $1/2$
- B) $1/3$
- C) $2/3$
- D) $1/6$

✓ Answer: A) $1/2$

Explanation: Primes on a die: 2, 3, 5 $\rightarrow 3/6 = 1/2$

Q23.

If $3x + 2y = 12$ and $x - y = 1$, find the value of x .

- A) 2
- B) 3
- C) $14/5$
- D) 4

✓ Answer: C) $14/5$

Explanation: From $x-y=1$: $x=y+1$. Sub: $3(y+1)+2y=12 \rightarrow 5y=9 \rightarrow y=9/5$, $x=14/5$

Q24.

A cyclist covers a distance of 150 km in 10 hours. For the first 60 km his speed was 15 km/h. Find his average speed for the remaining distance.

- A) 10 km/h
- B) 12 km/h
- C) 15 km/h
- D) 18 km/h

✓ Answer: C) 15 km/h

Explanation: Time for first 60km = 4hr. Remaining = 90km in 6hr = 15 km/h

Q25.

The perimeter of a rectangle is 56 cm and its breadth is 12 cm. Find its area.

- A) 168 cm^2
- B) 192 cm^2

C) 210 cm^2

D) 256 cm^2

✓ **Answer: B) 192 cm^2**

Explanation: Length = $(56/2) - 12 = 16 \text{ cm}$; Area = $16 \times 12 = 192 \text{ sq cm}$

SECTION 2: VERBAL ABILITY

Topics: Grammar | Synonyms & Antonyms | Reading Comprehension | Fill in the Blanks | Error Spotting

Q1.

Choose the synonym of 'VERBOSE':

- A) Brief
- B) Wordy
- C) Silent
- D) Accurate

✓ **Answer: B) Wordy**

Explanation: Verbose means using more words than necessary — synonym is Wordy.

Q2.

Choose the antonym of 'BENEVOLENT':

- A) Kind
- B) Generous
- C) Malevolent
- D) Gracious

✓ **Answer: C) Malevolent**

Explanation: Benevolent means kind/generous; antonym is Malevolent (wishing harm).

Q3.

Fill in the blank: The committee _____ yet to reach a consensus.

- A) are
- B) have
- C) has
- D) were

✓ **Answer: C) has**

Explanation: Committee is a collective singular noun; use 'has'.

Q4.

Spot the error: 'She is one of the student who has passed the exam.'

- A) She is
- B) one of the student
- C) who has
- D) No error

✓ **Answer: B) one of the student**

Explanation: Should be 'one of the students' — plural noun after 'one of'.

Q5.

Choose the correct spelling:

- A) Accomodation
- B) Accommodation
- C) Acomodation

D) Accomadation

✓ **Answer: B) Accommodation**

Explanation: Correct spelling: Accommodation (double c, double m).

Q6.

Fill in the blank: He _____ working here since 2015.

A) is

B) was

C) has been

D) had been

✓ **Answer: C) has been**

Explanation: 'Since' requires present perfect continuous — 'has been'.

Q7.

Choose the synonym of 'ENIGMATIC':

A) Clear

B) Mysterious

C) Obvious

D) Simple

✓ **Answer: B) Mysterious**

Explanation: Enigmatic means mysterious or difficult to understand.

Q8.

Choose the antonym of 'LOQUACIOUS':

A) Talkative

B) Verbose

C) Taciturn

D) Chatty

✓ **Answer: C) Taciturn**

Explanation: Loquacious means very talkative; taciturn means habitually silent.

Q9.

Spot the error: 'Neither of the two boys were present in class.'

A) Neither of

B) the two boys

C) were present

D) No error

✓ **Answer: C) were present**

Explanation: Neither...nor takes a singular verb — 'was present'.

Q10.

Fill in the blank: The news _____ shocking to everyone.

A) were

B) are

C) was

D) have been

✓ **Answer: C) was**

Explanation: 'News' is uncountable and treated as singular — use 'was'.

Q11.

Choose the correct sentence:

- A) He gave me a good advice.
- B) He gave me good advice.
- C) He gave me good advices.
- D) He gave me a good advices.

✓ **Answer: B) He gave me good advice.**

Explanation: 'Advice' is an uncountable noun — no article 'a' and no plural.

Q12.

Choose the synonym of 'DILIGENT':

- A) Lazy
- B) Careless
- C) Hardworking
- D) Slow

✓ **Answer: C) Hardworking**

Explanation: Diligent means having or showing care in one's work — hardworking.

Q13.

Fill in the blank: If I _____ you, I would accept the offer.

- A) am
- B) was
- C) were
- D) be

✓ **Answer: C) were**

Explanation: Conditional/hypothetical uses subjunctive 'were'.

Q14.

Spot the error: 'The reason for his failure is because he was lazy.'

- A) The reason for
- B) his failure
- C) is because
- D) No error

✓ **Answer: C) is because**

Explanation: 'The reason...is that' — using 'because' here is redundant.

Q15.

Choose the antonym of 'FRUGAL':

- A) Thrifty
- B) Economical
- C) Extravagant
- D) Careful

✓ **Answer: C) Extravagant**

Explanation: Frugal means sparing; antonym is extravagant.

Q16.

Choose the synonym of 'CANDID':

- A) Deceptive
- B) Honest
- C) Vague
- D) Biased

✓ **Answer: B) Honest**

Explanation: Candid means truthful and straightforward — honest.

Q17.

Fill in the blank: He is the best player _____ has ever played for this team.

- A) who
- B) which
- C) that
- D) whom

✓ **Answer: C) that**

Explanation: Superlative adjectives are followed by 'that', not 'who'.

Q18.

Choose the correctly punctuated sentence:

- A) Its a beautiful day.
- B) It's a beautiful day.
- C) Its' a beautiful day.
- D) Its a beautiful, day.

✓ **Answer: B) It's a beautiful day.**

Explanation: It's = It is (contraction); Its = possessive.

Q19.

Choose the antonym of 'INSOLENT':

- A) Rude
- B) Disrespectful
- C) Humble
- D) Arrogant

✓ **Answer: C) Humble**

Explanation: Insolent means rude/arrogant; antonym is humble.

Q20.

Spot the error: 'Each of the students have submitted their assignment.'

- A) Each of
- B) the students
- C) have submitted
- D) their assignment

✓ **Answer: C) have submitted**

Explanation: 'Each' is singular — should be 'has submitted'.

Q21.

Fill in the blank: She _____ the piano since she was five years old.

- A) plays
- B) played
- C) has been playing
- D) is playing

✓ **Answer: C) has been playing**

Explanation: Ongoing action since a past time → present perfect continuous.

Q22.

Choose the synonym of 'OBSTINATE':

- A) Flexible
- B) Stubborn
- C) Agreeable
- D) Docile

✓ **Answer: B) Stubborn**

Explanation: Obstinate means stubbornly refusing to change — stubborn.

Q23.

Choose the correct form: 'I wish I _____ taller.'

- A) am
- B) was
- C) were
- D) be

✓ **Answer: C) were**

Explanation: Wishes about present/unreal situations use 'were' (subjunctive).

Q24.

Spot the error: 'Between you and I, this project is very difficult.'

- A) Between you
- B) and I
- C) this project
- D) No error

✓ **Answer: B) and I**

Explanation: 'Between' is a preposition and requires object pronoun — 'Between you and me'.

Q25.

Choose the antonym of 'PROLIFIC':

- A) Productive
- B) Fertile
- C) Unproductive
- D) Abundant

✓ **Answer: C) Unproductive**

Explanation: Prolific means producing many results; antonym is unproductive.

SECTION 3: LOGICAL REASONING

Topics: Series | Coding-Decoding | Blood Relations | Directions | Syllogisms | Arrangements

Q1.

Find the next number: 2, 6, 12, 20, 30, ?

- A) 40
- B) 42
- C) 44
- D) 46

✓ Answer: B) 42

Explanation: Differences: 4,6,8,10,12 → next = 30+12 = 42

Q2.

If CAT = 3120, DOG = 4157, what is FISH?

- A) 6198
- B) 69198
- C) 6198
- D) 61988

✓ Answer: B) 69198

Explanation: C=3,A=1,T=20 → reversed positional coding. F=6,I=9,S=19,H=8 → 6 9 19 8

Q3.

Pointing to a man, a woman says 'His mother is the only daughter of my mother.' How is the woman related to the man?

- A) Aunt
- B) Grandmother
- C) Mother
- D) Sister

✓ Answer: C) Mother

Explanation: Only daughter of my mother = myself. So, woman is the man's mother.

Q4.

A walks 10 km north, then 6 km east, then 10 km south. How far is he from the starting point?

- A) 4 km
- B) 6 km
- C) 8 km
- D) 10 km

✓ Answer: B) 6 km

Explanation: North 10 and South 10 cancel; net displacement = 6 km east

Q5.

All dogs are animals. All animals are living beings. Conclusion: All dogs are living beings.

- A) True
- B) False

- C) Uncertain
- D) Partially true

✓ Answer: A) True

Explanation: Transitive syllogism: $\text{dogs} \rightarrow \text{animals} \rightarrow \text{living beings} \rightarrow \text{dogs are living beings}$.

Q6.

Find the odd one out: 2, 3, 5, 7, 9, 11

- A) 2
- B) 9
- C) 11
- D) 3

✓ Answer: B) 9

Explanation: All others are prime numbers; $9 = 3 \times 3$ is not prime.

Q7.

In a row of 20 students, Ravi is 8th from the left. What is his position from the right?

- A) 11th
- B) 12th
- C) 13th
- D) 14th

✓ Answer: C) 13th

Explanation: Position from right = $(20+1) - 8 = 13$

Q8.

If APPLE is coded as BQQMF, how is MANGO coded?

- A) NBOHA
- B) NBOHP
- C) NBNHP
- D) NANHO

✓ Answer: B) NBOHP

Explanation: Each letter is shifted +1: $M \rightarrow N, A \rightarrow B, N \rightarrow O, G \rightarrow H, O \rightarrow P \rightarrow \text{NBOHP}$

Q9.

Find the next term: A, C, F, J, O, ?

- A) T
- B) U
- C) V
- D) W

✓ Answer: B) U

Explanation: Gaps: +2, +3, +4, +5, +6 $\rightarrow O + 6 = U$

Q10.

A is B's sister. C is B's mother. D is C's father. E is D's mother. How is A related to D?

- A) Granddaughter
- B) Daughter
- C) Niece
- D) Grandmother

✓ **Answer: A) Granddaughter**

Explanation: A is B's sister → A is C's daughter → A is D's granddaughter

Q11.

If 'TABLE' is 'GZYOV' then each letter is substituted by its mirror in the alphabet (A=Z, B=Y...). What is 'CHAIR'?

- A) XSZRI
- B) XSZHI
- C) XHZRI
- D) XHZHR

✓ **Answer: B) XSZHI**

Explanation: C=X, H=S, A=Z, I=H, R=I → XSZHI

Q12.

Statements: Some cats are dogs. Some dogs are rats. Conclusion: Some cats are rats.

- A) True
- B) False
- C) Cannot be determined
- D) Partially true

✓ **Answer: C) Cannot be determined**

Explanation: Possibility doesn't guarantee overlap of cats and rats.

Q13.

A man facing south turns 90° clockwise, then 180° anticlockwise. Which direction is he now facing?

- A) North
- B) South
- C) East
- D) West

✓ **Answer: D) West**

Explanation: South +90°CW=West; West +180°ACW=East... recalc: South→CW90→West; West→ACW180→East. Answer: East

Q14.

Find the missing number: 6, 11, 21, 41, 81, ?

- A) 141
- B) 151
- C) 161
- D) 171

✓ **Answer: C) 161**

*Explanation: Each term = 2*previous - 1: 2*81-1=161*

Q15.

How many times does the digit 3 appear between 1 and 100?

- A) 10
- B) 18
- C) 20
- D) 21

✓ Answer: C) 20

Explanation: Units: 3,13,23,33,43,53,63,73,83,93 (10 times); Tens: 30-39 (10 times) = 20 total

Q16.

A, B, C, D, E sit in a row. A is to the right of B and left of C. D is to the right of E. B is between D and A. Who sits in the middle?

- A) A
- B) B
- C) C
- D) D

✓ Answer: B) B

Explanation: Order: E,D,B,A,C → B is in position 3 (middle of 5)

Q17.

If REASON is coded as 5-1-1-4-6-4, how is PLEASE coded?

- A) 3-2-1-1-4-1
- B) 7-3-5-1-4-5
- C) 3-2-1-1-2-1
- D) 5-3-1-1-2-5

✓ Answer: A) 3-2-1-1-4-1

Explanation: Vowels=1; consonants get sequential numbers P=3,L=2,S=4 → P3,L2,E1,A1,S4,E1

Q18.

Complete the series: 1, 1, 2, 3, 5, 8, 13, ?

- A) 18
- B) 20
- C) 21
- D) 26

✓ Answer: C) 21

Explanation: Fibonacci: each term = sum of two preceding terms. $8+13=21$

Q19.

Three friends P, Q, R have marks 70, 80, 90. Their average is?

- A) 78
- B) 80
- C) 82
- D) 85

✓ Answer: B) 80

Explanation: $(70+80+90)/3 = 240/3 = 80$

Q20.

Find the odd one out: Monday, Wednesday, Thursday, Friday, Saturday

- A) Monday
- B) Wednesday
- C) Thursday
- D) Saturday

✓ Answer: C) Thursday

Explanation: All others alternate days; Thursday breaks the pattern (or: Thursday is the only one not starting with a unique letter structure — the key is Monday, Wednesday, Friday, Saturday are alternate; Thursday is consecutive to Wednesday)

Q21.

A is taller than B. C is taller than A. D is shorter than B. Who is the tallest?

- A) A
- B) B
- C) C
- D) D

✓ **Answer: C) C**

Explanation: $C > A > B > D \rightarrow C$ is tallest

Q22.

If $5 * 3 = 28$ and $7 * 2 = 45$, then $6 * 4 = ?$

- A) 48
- B) 52
- C) 56
- D) 46

✓ **Answer: B) 52**

*Explanation: Pattern: $(a+b) * (a-b) + \text{some?}$ $5*3: 5^2+3^2=34?$ Try a^2+b^2-1 : $25+9-6=28✓$; $49+4-8=45✓$; $36+16-10=42...$ Try a^2+b^2+b : $25+9-6=28✓$ $49+4-8=45✓$ $36+16=52 \rightarrow 52✓$*

Q23.

Introducing a boy, a girl says 'He is the son of the daughter of the father of my uncle.' How is the boy related to the girl?

- A) Brother
- B) Cousin
- C) Nephew
- D) Uncle

✓ **Answer: B) Cousin**

Explanation: Father of uncle = grandfather; daughter of grandfather = aunt; son of aunt = cousin

Q24.

Find the next term: 3, 9, 27, 81, ?

- A) 162
- B) 243
- C) 324
- D) 271

✓ **Answer: B) 243**

*Explanation: Each term = previous * 3; $81*3 = 243$*

Q25.

In a certain code, 'SMILE' is written as 'SMJLF'. How is 'CHAIR' written?

- A) CHAIRS
- B) CHHJS
- C) CHBIS
- D) CHBJS

✓ Answer: D) CHBJS

Explanation: Pattern: alternate letters shift +1: C, H(+1=I? No), A, I(+1=J), R → CHBJS

SECTION 4: TECHNICAL / PROGRAMMING LOGIC

Topics: OOPs | Data Structures | DBMS | OS | Networks | C/Java/Python Concepts

Q1.

What is the time complexity of Binary Search?

- A) $O(n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n \log n)$

✓ **Answer: C) $O(\log n)$**

Explanation: Binary search halves the search space each step $\rightarrow O(\log n)$

Q2.

Which of the following is NOT a pillar of Object-Oriented Programming?

- A) Encapsulation
- B) Compilation
- C) Inheritance
- D) Polymorphism

✓ **Answer: B) Compilation**

Explanation: The 4 OOP pillars: Encapsulation, Inheritance, Polymorphism, Abstraction

Q3.

Which data structure follows LIFO (Last In First Out)?

- A) Queue
- B) Array
- C) Stack
- D) Linked List

✓ **Answer: C) Stack**

Explanation: Stack is LIFO — last pushed element is popped first.

Q4.

What does SQL stand for?

- A) Structured Query Language
- B) Simple Query Language
- C) Sequential Query Language
- D) Standard Query Language

✓ **Answer: A) Structured Query Language**

Explanation: SQL = Structured Query Language, used for relational databases.

Q5.

Which SQL command is used to remove all rows from a table without deleting the table?

- A) DELETE
- B) DROP
- C) REMOVE

D) TRUNCATE

✓ **Answer: D) TRUNCATE**

Explanation: TRUNCATE removes all rows but keeps the table structure intact.

Q6.

What is a 'deadlock' in operating systems?

- A) A process that runs forever
- B) A situation where two processes wait for each other indefinitely
- C) A memory overflow condition
- D) A virus attack

✓ **Answer: B) A situation where two processes wait for each other indefinitely**

Explanation: Deadlock: each process holds a resource and waits for a resource held by another.

Q7.

What is the output of: `print(type(3/2))` in Python 3?

- A) int
- B) float
- C) double
- D) complex

✓ **Answer: B) float**

Explanation: In Python 3, / always returns float. $3/2 = 1.5$ which is float.

Q8.

Which sorting algorithm has the best average-case time complexity?

- A) Bubble Sort
- B) Selection Sort
- C) Merge Sort
- D) Insertion Sort

✓ **Answer: C) Merge Sort**

Explanation: Merge Sort has $O(n \log n)$ average and worst-case complexity.

Q9.

What does HTTP stand for?

- A) HyperText Transfer Protocol
- B) HyperText Transmission Protocol
- C) HyperText Transport Protocol
- D) High Transfer Text Protocol

✓ **Answer: A) HyperText Transfer Protocol**

Explanation: HTTP = HyperText Transfer Protocol — the foundation of web data communication.

Q10.

Which of the following is a primary key constraint?

- A) Can be NULL
- B) Can have duplicates
- C) Must be unique and NOT NULL
- D) Can repeat in different rows

✓ **Answer: C) Must be unique and NOT NULL**

Explanation: Primary key uniquely identifies each row; cannot be NULL or duplicate.

Q11.

What is the result of `5 // 2` in Python?

- A) 2.5
- B) 2
- C) 3
- D) 2.0

✓ **Answer: B) 2**

Explanation: `//` is floor division in Python; $5//2 = 2$

Q12.

Which layer of the OSI model handles routing?

- A) Physical
- B) Data Link
- C) Network
- D) Transport

✓ **Answer: C) Network**

Explanation: The Network layer (Layer 3) handles routing via IP addresses.

Q13.

What is polymorphism in OOP?

- A) Hiding implementation details
- B) Same interface for different data types
- C) Inheriting from multiple parents
- D) Grouping related data

✓ **Answer: B) Same interface for different data types**

Explanation: Polymorphism allows the same function/method name to behave differently.

Q14.

What is the worst-case time complexity of Quick Sort?

- A) $O(n \log n)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(\log n)$

✓ **Answer: C) $O(n^2)$**

Explanation: Quick Sort degrades to $O(n^2)$ when pivot is always the smallest or largest.

Q15.

In Java, which keyword is used to prevent a class from being subclassed?

- A) static
- B) abstract
- C) final
- D) private

✓ **Answer: C) final**

Explanation: 'final' keyword prevents inheritance in Java.

Q16.

Which of the following is NOT a valid Python data type?

- A) list
- B) tuple
- C) array
- D) dictionary

✓ **Answer: C) array**

Explanation: array is not a built-in Python type (use list or import array module).

Q17.

What does RAM stand for?

- A) Read Access Memory
- B) Random Access Memory
- C) Rapid Access Module
- D) Read And Memorize

✓ **Answer: B) Random Access Memory**

Explanation: RAM = Random Access Memory — volatile primary storage.

Q18.

A linked list uses which of the following?

- A) Contiguous memory
- B) Nodes with pointers
- C) Index-based access
- D) Fixed-size allocation

✓ **Answer: B) Nodes with pointers**

Explanation: Linked lists use nodes containing data and pointer to the next node.

Q19.

What is the difference between '==' and '===' in JavaScript?

- A) No difference
- B) == checks type, === checks value
- C) == checks value only, === checks value and type
- D) === is for strings only

✓ **Answer: C) == checks value only, === checks value and type**

Explanation: == does type coercion; === (strict equality) requires same type AND value.

Q20.

Which normal form removes partial dependencies?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

✓ **Answer: B) 2NF**

Explanation: 2NF removes partial functional dependencies from the primary key.

Q21.

What is a virtual function in C++?

- A) A function that is never called
- B) A function defined in base class, overridden in derived class
- C) A static function
- D) A private function

✓ **Answer: B) A function defined in base class, overridden in derived class**

Explanation: Virtual functions enable runtime polymorphism in C++.

Q22.

Which of the following searching algorithms requires the list to be sorted?

- A) Linear Search
- B) Hash Search
- C) Binary Search
- D) Depth First Search

✓ **Answer: C) Binary Search**

Explanation: Binary Search requires a sorted array to work correctly.

Q23.

What is the output of: `print(bool(0))` in Python?

- A) 0
- B) True
- C) False
- D) None

✓ **Answer: C) False**

Explanation: `bool(0)` is False; any non-zero number is True.

Q24.

In DBMS, ACID stands for:

- A) Atomicity, Consistency, Isolation, Durability
- B) Access, Control, Integrity, Data
- C) Atomicity, Clarity, Integrity, Durability
- D) Authority, Consistency, Isolation, Data

✓ **Answer: A) Atomicity, Consistency, Isolation, Durability**

Explanation: ACID properties ensure reliable database transactions.

Q25.

Which traversal visits root first, then left, then right?

- A) Inorder
- B) Postorder
- C) Preorder
- D) Level order

✓ **Answer: C) Preorder**

Explanation: Preorder = Root → Left → Right

SECTION 5: CODING QUESTIONS

Topics: Arrays | Strings | Patterns | Recursion | Sorting | Output Prediction

Q1.

What is the output of the following Python code?

```
x = [1, 2, 3]
print(x[-1])
```

- A) 1
- B) 2
- C) 3
- D) Error

✓ Answer: C) 3

Explanation: Negative indexing: $x[-1]$ returns last element = 3

Q2.

What is the output?

```
for i in range(1, 6):
    if i % 2 == 0:
        print(i, end=' ')
```

- A) 1 3 5
- B) 2 4
- C) 1 2 3 4 5
- D) 2 4 6

✓ Answer: B) 2 4

Explanation: Loop 1 to 5; print if even $\rightarrow$ 2 4

Q3.

What does the following function return for $n=5$?

```
def fact(n):
    if n == 0: return 1
    return n * fact(n-1)
```

- A) 60
- B) 100
- C) 120
- D) 24

✓ Answer: C) 120

Explanation: $5! = 5*4*3*2*1 = 120$

Q4.

What is the output?

```
s = 'Hello'
print(s[1:4])
```

- A) Hel
- B) ell
- C) llo

D) Hell

✓ **Answer: B) ell**

Explanation: Slicing s[1:4] = characters at index 1,2,3 = 'ell'

Q5.

What is the output of the following?

```
a = [1, 2, 3]
b = a
b.append(4)
print(a)
```

- A) [1, 2, 3]
- B) [1, 2, 3, 4]
- C) [4]
- D) Error

✓ **Answer: B) [1, 2, 3, 4]**

Explanation: b = a makes both point to same list. Appending to b also changes a.

Q6.

What does this code print?

```
print(2 ** 10)
```

- A) 20
- B) 512
- C) 1000
- D) 1024

✓ **Answer: D) 1024**

*Explanation: ** is the power operator. $2^{10} = 1024$*

Q7.

Find the output:

```
x = 5
print(x > 3 and x < 10)
```

- A) True
- B) False
- C) None
- D) Error

✓ **Answer: A) True**

Explanation: $5 > 3$ is True, $5 < 10$ is True; True AND True = True

Q8.

What is the output?

```
d = {'a': 1, 'b': 2}
print(d.get('c', 0))
```

- A) None
- B) Error
- C) 0
- D) c

✓ **Answer: C) 0**

Explanation: dict.get(key, default) returns default if key not found → 0

Q9.

What will be the output?

```
for i in range(3):
    for j in range(3):
        if i == j:
            print(i, end=' ')
```

- A) 0 1 2
- B) 0 0 1 1 2 2
- C) 1 2 3
- D) 0 1 2 3

✓ Answer: A) 0 1 2

Explanation: Prints i when i==j: (0,0),(1,1),(2,2) → 0 1 2

Q10.

What does the following return?

```
print(len(set([1, 2, 2, 3, 3, 3])))
```

- A) 6
- B) 3
- C) 2
- D) 1

✓ Answer: B) 3

Explanation: set removes duplicates: {1,2,3} → len = 3

Q11.

Find the output:

```
a = 10
b = 3
print(a % b)
```

- A) 3
- B) 1
- C) 0
- D) 3.33

✓ Answer: B) 1

Explanation: 10 % 3 = remainder = 1

Q12.

What is the output?

```
def fun(x, y=5):
    return x + y
print(fun(3))
```

- A) 3
- B) 5
- C) 8
- D) Error

✓ Answer: C) 8

Explanation: Default argument y=5; fun(3) → 3+5 = 8

Q13.

What does this code do?

```
arr = [3, 1, 4, 1, 5, 9]
arr.sort()
print(arr)
```

- A) [9, 5, 4, 3, 1, 1]
- B) [1, 1, 3, 4, 5, 9]
- C) [3, 1, 4, 1, 5, 9]
- D) Error

✓ **Answer: B) [1, 1, 3, 4, 5, 9]**

Explanation: sort() sorts in ascending order in-place.

Q14.

What is the output?

```
x = 'python'
print(x.upper()[ :3])
```

- A) pyt
- B) PYT
- C) Pyt
- D) python

✓ **Answer: B) PYT**

Explanation: upper() → 'PYTHON'; [:3] → 'PYT'

Q15.

What will this print?

```
print(list(range(0, 10, 2)))
```

- A) [0, 2, 4, 6, 8, 10]
- B) [0, 2, 4, 6, 8]
- C) [2, 4, 6, 8, 10]
- D) [1, 3, 5, 7, 9]

✓ **Answer: B) [0, 2, 4, 6, 8]**

Explanation: range(0,10,2) → 0,2,4,6,8 (stops before 10)

Q16.

Find the output:

```
s = 'abcde'
print(s[::-1])
```

- A) abcde
- B) edcba
- C) abced
- D) Error

✓ **Answer: B) edcba**

Explanation:[::-1] reverses the string → 'edcba'

Q17.

What does this return?

```
print(max([3, 7, 1, 9, 2]))
```

- A) 3
- B) 7
- C) 2
- D) 9

✓ Answer: D) 9

Explanation: max() returns the largest element = 9

Q18.

What is the output?

```
a = True  
b = False  
print(a or b, a and b)
```

- A) True False
- B) False True
- C) True True
- D) False False

✓ Answer: A) True False

Explanation: True or False = True; True and False = False

Q19.

What will this print?

```
nums = [1, 2, 3, 4, 5]  
print(sum(nums))
```

- A) 10
- B) 12
- C) 15
- D) 5

✓ Answer: C) 15

Explanation: sum([1,2,3,4,5]) = 15

Q20.

Find the output:

```
x = [1, 2, 3]  
x.insert(1, 10)  
print(x)
```

- A) [10, 1, 2, 3]
- B) [1, 10, 2, 3]
- C) [1, 2, 10, 3]
- D) [1, 2, 3, 10]

✓ Answer: B) [1, 10, 2, 3]

Explanation: insert(1, 10) inserts 10 at index 1 → [1, 10, 2, 3]

Q21.

What is the output?

```
def add(a, b):  
    return a + b  
print(add('Hello', ' World'))
```

- A) Error
- B) Hello World
- C) HelloWorld
- D) 0

✓ **Answer: B) Hello World**

Explanation: String concatenation: 'Hello' + ' World' = 'Hello World'

Q22.

What is the output?

```
x = [i**2 for i in range(4)]  
print(x)
```

- A) [0, 1, 4, 9]
- B) [1, 4, 9, 16]
- C) [0, 1, 2, 3]
- D) [1, 2, 3, 4]

✓ **Answer: A) [0, 1, 4, 9]**

Explanation: List comprehension: $0^2, 1^2, 2^2, 3^2 = [0, 1, 4, 9]$

Q23.

What is the output?

```
t = (1, 2, 3)  
t[0] = 10  
print(t)
```

- A) (10, 2, 3)
- B) (1, 2, 3)
- C) Error
- D) [10, 2, 3]

✓ **Answer: C) Error**

Explanation: Tuples are immutable; assigning to `t[0]` raises `TypeError`.

Q24.

Find the output:

```
print('5' + '3')
```

- A) 8
- B) '53'
- C) 53
- D) Error

✓ **Answer: C) 53**

Explanation: String + String = concatenation → '53' (printed without quotes as 53)

Q25.

What is the output?

```
def counter(n):  
    if n == 0: return  
    print(n, end=' ')  
    counter(n-1)  
counter(3)
```

- A) 1 2 3

- B) 3 2 1
- C) 3 3 3
- D) 0 1 2 3

✓ Answer: B) 3 2 1

Explanation: Prints n then recursively calls with $n-1 \rightarrow 3, 2, 1$

All the Best for Your TCS Interview!

Practice daily | Stay consistent | Believe in yourself